Quantum Computing for Climate and Sustainability

Abstract

Climate change research involves processing large and complex datasets to understand climate patterns, predict environmental changes, and support sustainable decision-making. However, conventional computing approaches can face limitations when handling highly complex climate models, optimization problems, and large-scale simulations. This creates a need to explore emerging computational technologies that can address these challenges.

Quantum computing is being explored as a potential approach for solving complex computational problems more efficiently. Its ability to perform quantum parallelism and handle optimization and simulation problems provides opportunities for applications in climate modeling, renewable energy optimization, carbon reduction, and environmental sustainability. Therefore, studying existing research on quantum computing can help identify its potential contribution to climate change research. **Existing Strategies and Techniques**

- Classical high-performance computing for climate modeling
- Artificial intelligence and machine learning for climate prediction
- Quantum optimization algorithms for energy and resource management
- Quantum simulation techniques for complex environmental models
- Hybrid classical–quantum computing approaches

Proposed Solution

This study proposes a knowledge-mapping approach to analyze existing research on the applications of quantum computing in climate change research and sustainable innovation. Approximately ten research papers will be systematically reviewed and compared based on their objectives, quantum techniques, applications, findings, limitations, and future research directions. The study will identify major research trends, existing gaps, and potential areas where quantum computing can support climate and sustainability-related applications.

The expected results are a structured understanding of the current state of quantum computing research in climate and sustainability. The study is expected to identify important applications, commonly used quantum techniques, major challenges, and research gaps. The findings can provide future researchers with a clear overview of how quantum computing may contribute to climate modeling, energy optimization, and sustainable development.

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